BackProp Vs Predictive Coding

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What is PC?

Predictive coding(PC) is a neural computation principle where each layer tries to **predict the activity of another layer**, and the difference between prediction and actual activity is called a **prediction error**.

The central idea is: The network updates its hidden states to reduce prediction error.

1. Arbitrary Graph NN with Backprop

Backprop Vs PC

Scenario: customer loan eligible amount prediction

1. customer loan approved data

Age	Salary	Marital_Status	Sex	Target Loan
30	38000	married	male	150000
25	25000	not married	female	80000
40	15000	married	female	30000

2. After categorical to numeric conversion

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
30	38000	1	0	1	0	150000
25	25000	0	1	0	1	80000
40	15000	1	0	0	1	30000

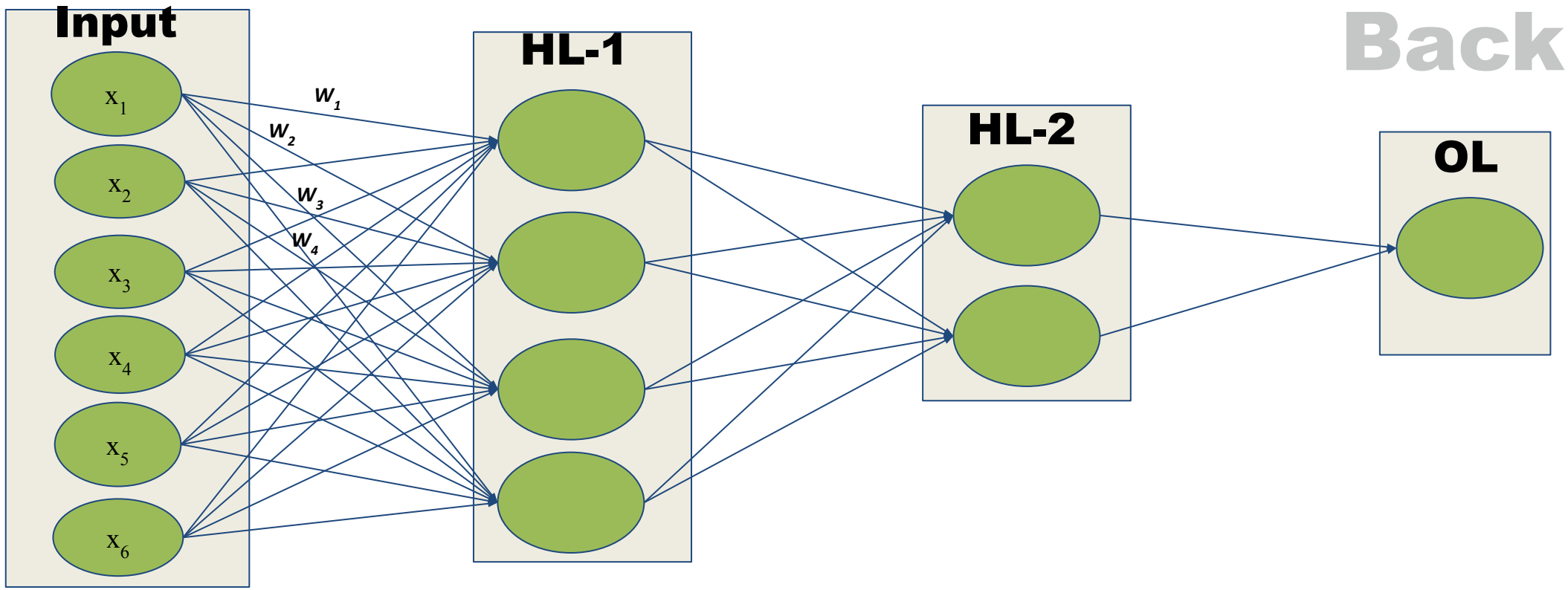
3. Data after Min-Max Scaling

How do we train?

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Backprop

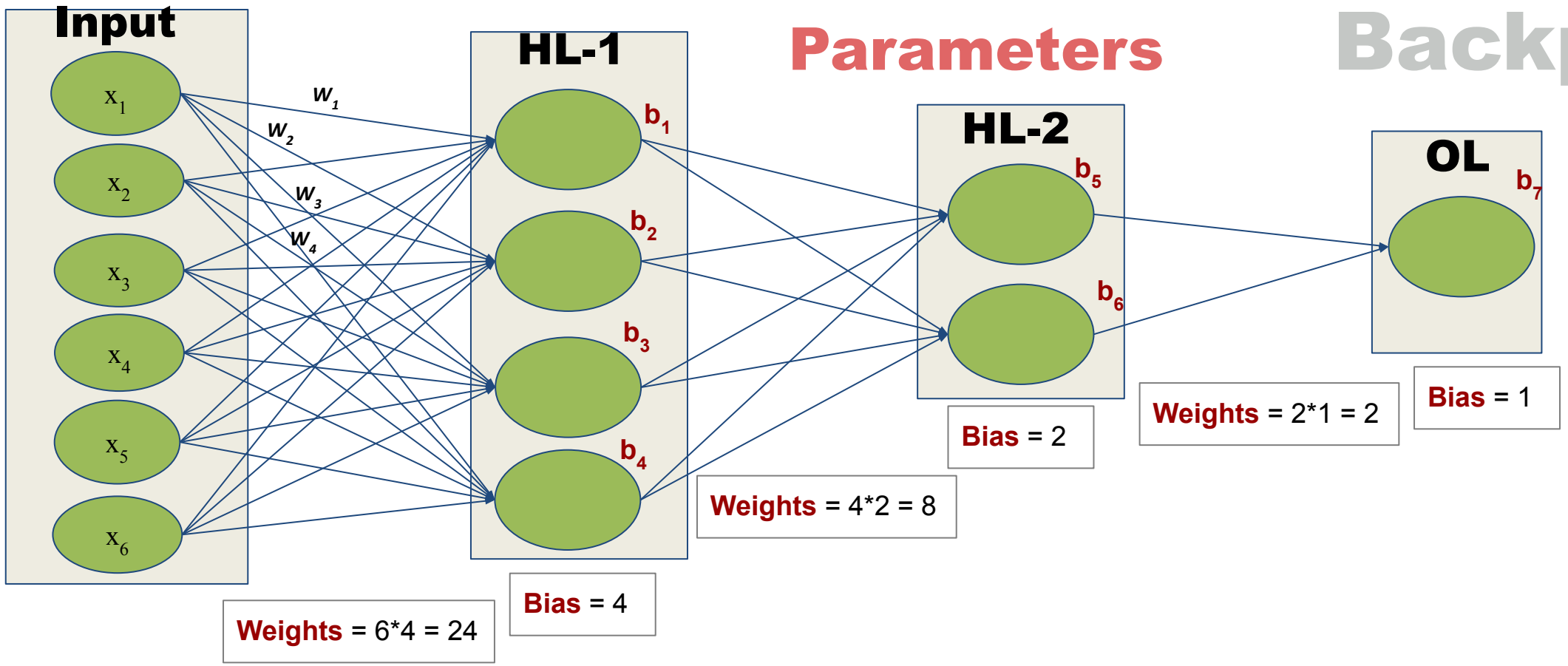


Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Parameters

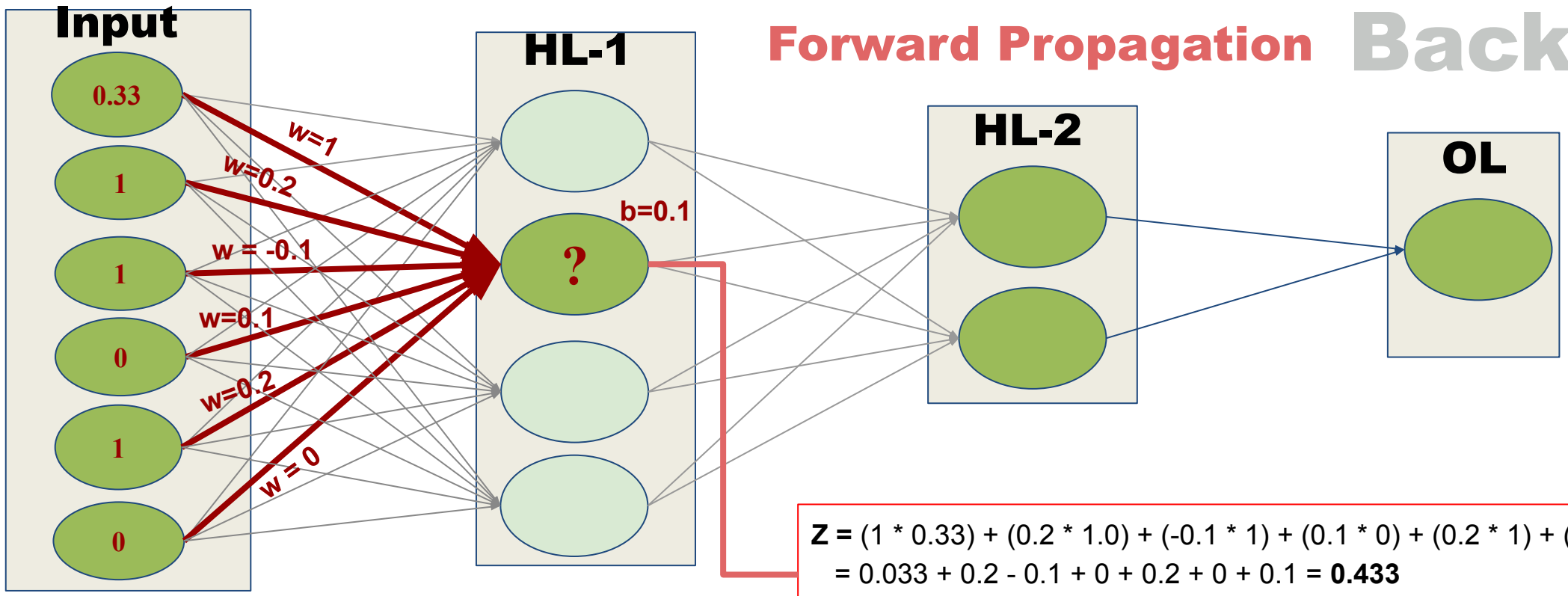
Backprop



Total Weights = Number of connections = 34
Total Bias = Number of neurons in **Hidden layers** and **output layer** = 4 + 2 + 1= 7
Total Parameters = 34 + 7 = 41

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Forward Propagation Backprop

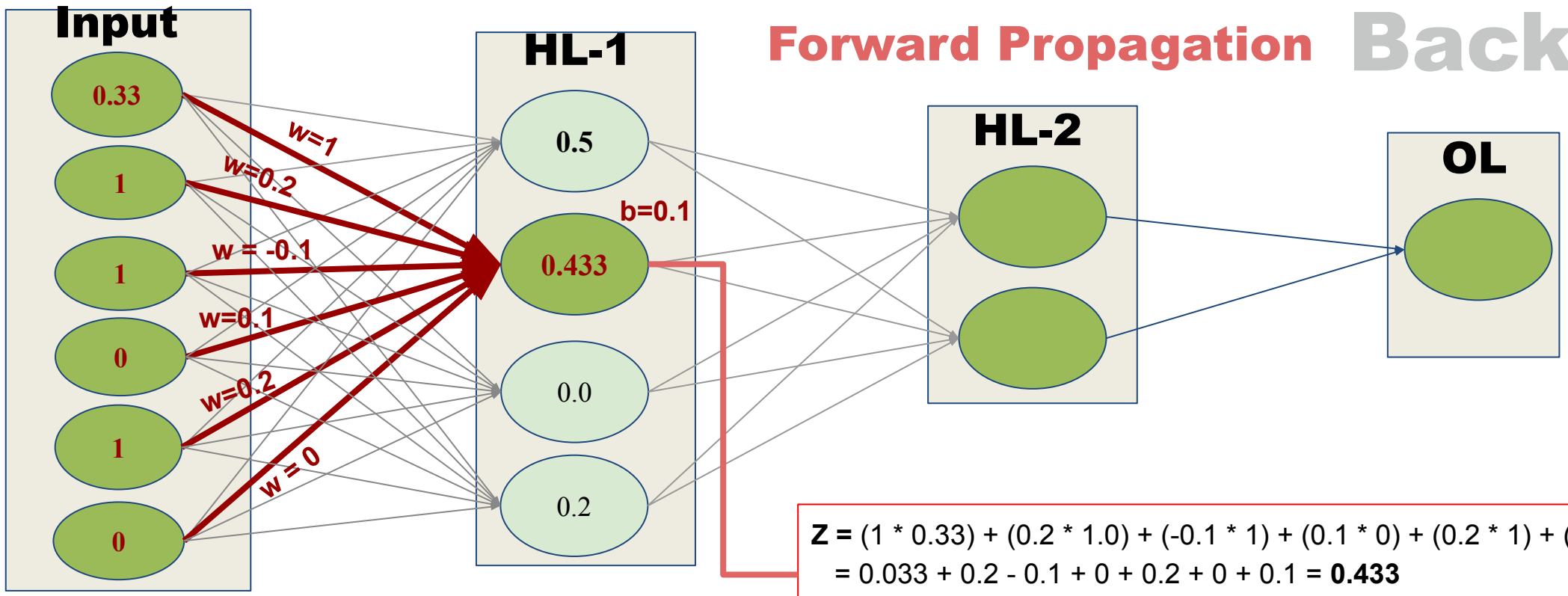
Z = (1 * 0.33) + (0.2 * 1.0) + (-0.1 * 1) + (0.1 * 0) + (0.2 * 1) + (0 * 0) + **0.1**
= 0.033 + 0.2 - 0.1 + 0 + 0.2 + 0 + 0.1 = **0.433**
A = Relu(Z) = Relu(0.433) **= 0.433**

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Forward Propagation

Backprop

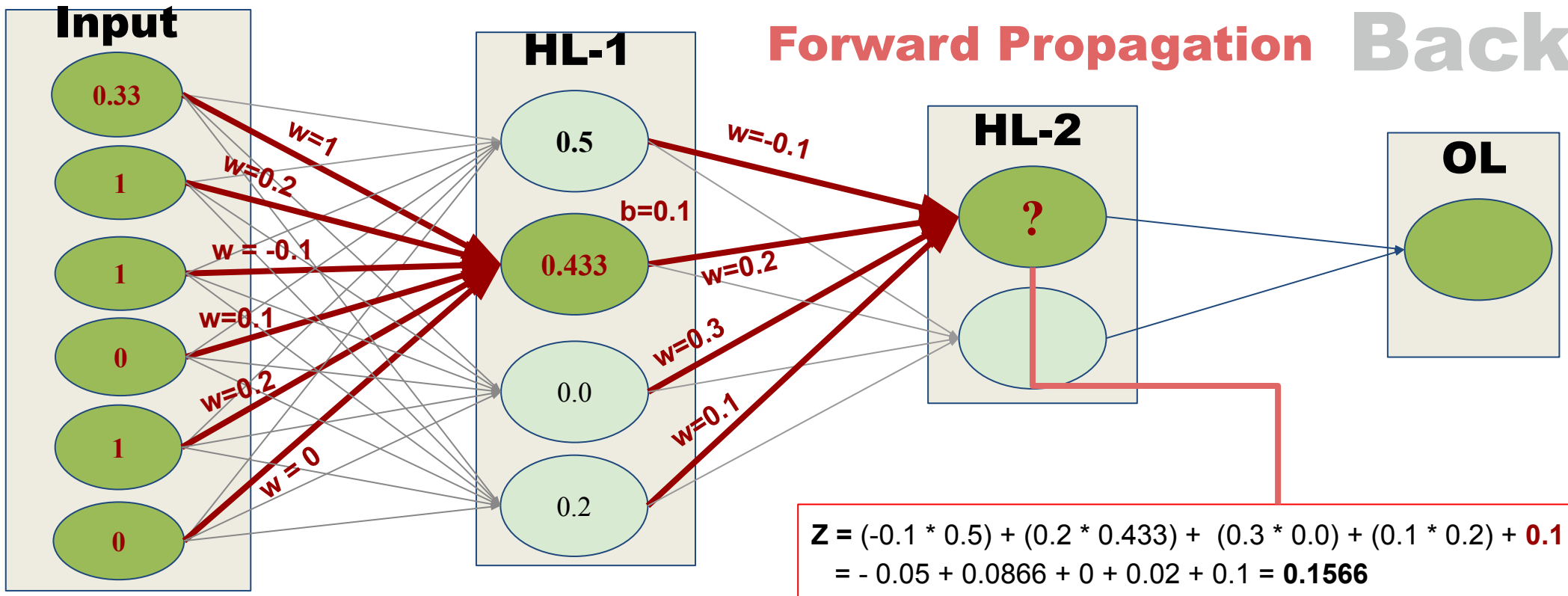
Z = (1 * 0.33) + (0.2 * 1.0) + (-0.1 * 1) + (0.1 * 0) + (0.2 * 1) + (0 * 0) + **0.1**
= 0.033 + 0.2 - 0.1 + 0 + 0.2 + 0 + 0.1 = **0.433**
A = Relu(Z) = Relu(0.433) **= 0.433**

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Forward Propagation Backprop

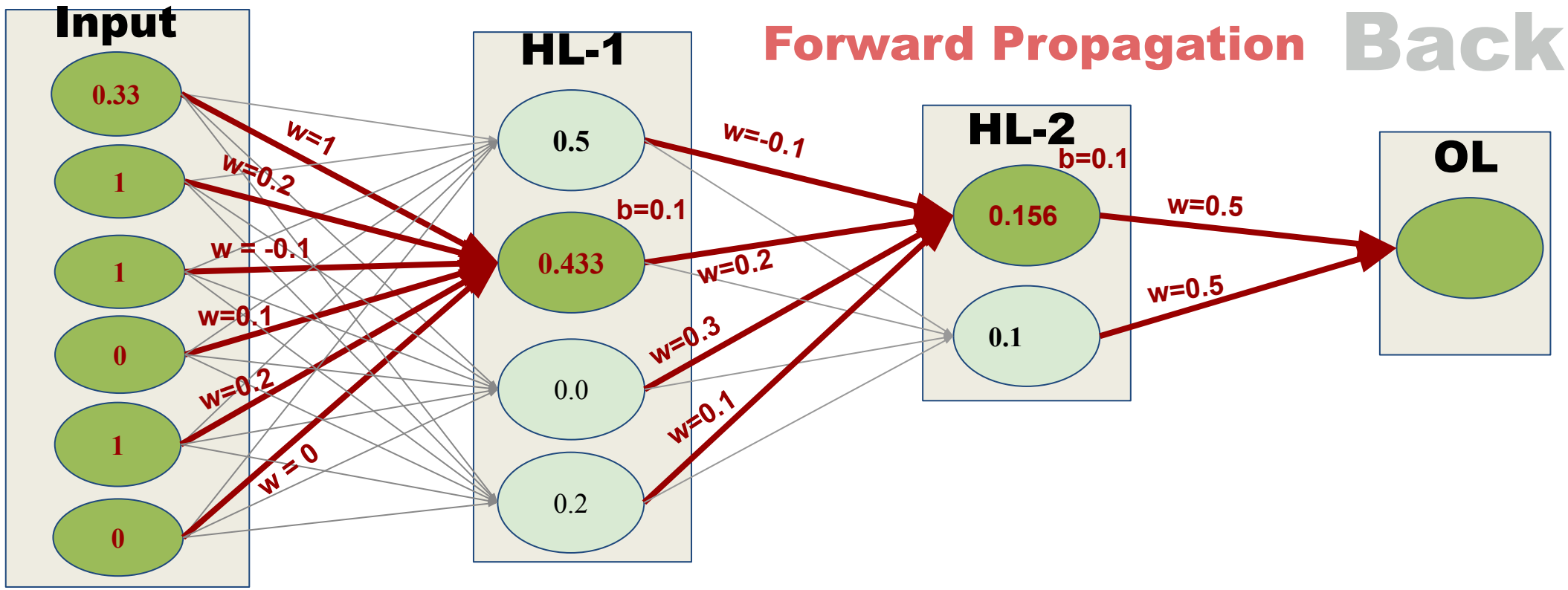
Z = (-0.1 * 0.5) + (0.2 * 0.433) + (0.3 * 0.0) + (0.1 * 0.2) + **0.1**
= - 0.05 + 0.0866 + 0 + 0.02 + 0.1 = **0.1566**
A = Relu(Z) = Relu(0.1566) **= 0.1566**

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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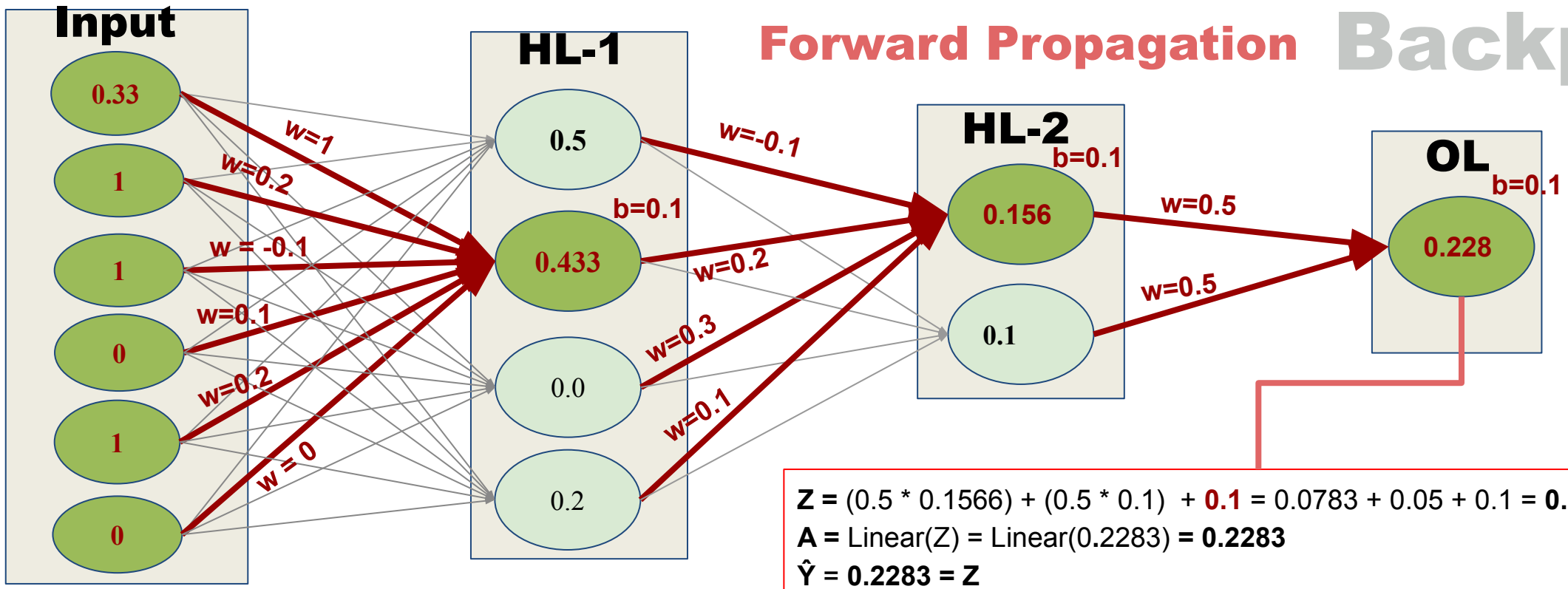
Forward Propagation Backprop

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Forward Propagation Backprop

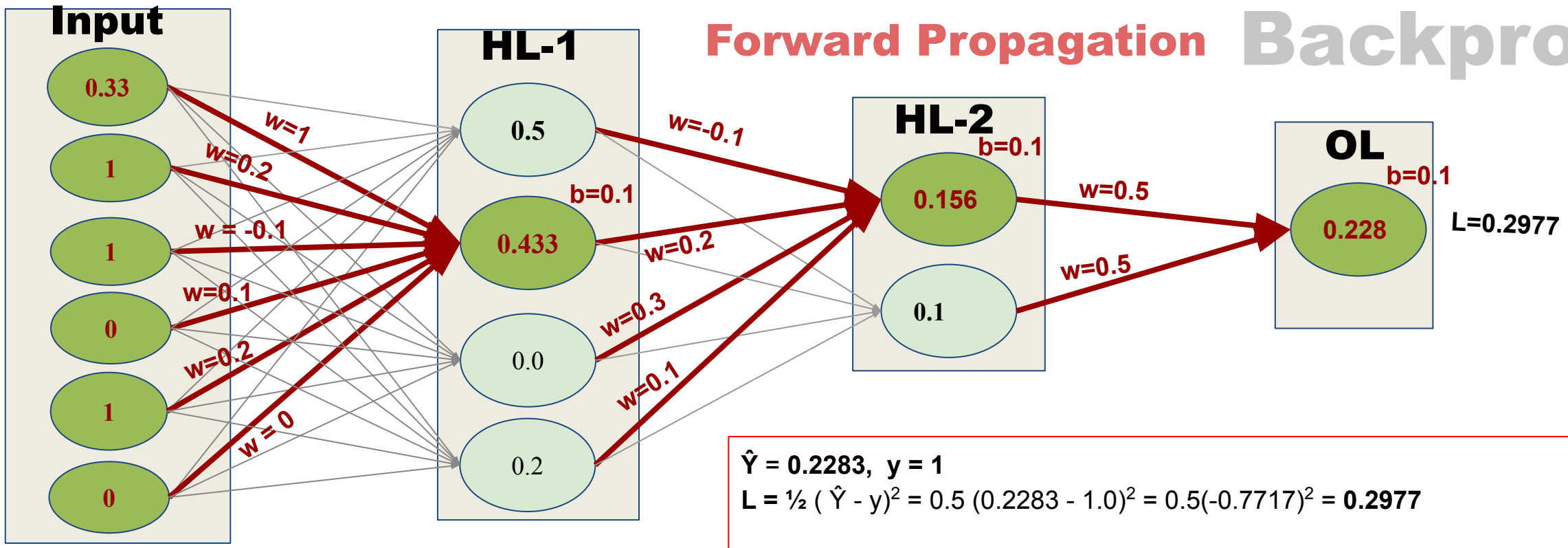
Z = (0.5 * 0.1566) + (0.5 * 0.1) + **0.1** = 0.0783 + 0.05 + 0.1 = **0.2283**
A = Linear(Z) = Linear(0.2283) **= 0.2283**
Ŷ = 0.2283 = Z

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Forward Propagation Backprop

$\hat{Y}$ = **0.2283, y = 1**

L = ½ ($\hat{Y}$ - y)^2 = 0.5 (0.2283 - 1.0)^2 = 0.5(-0.7717)^2 = **0.2977**

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

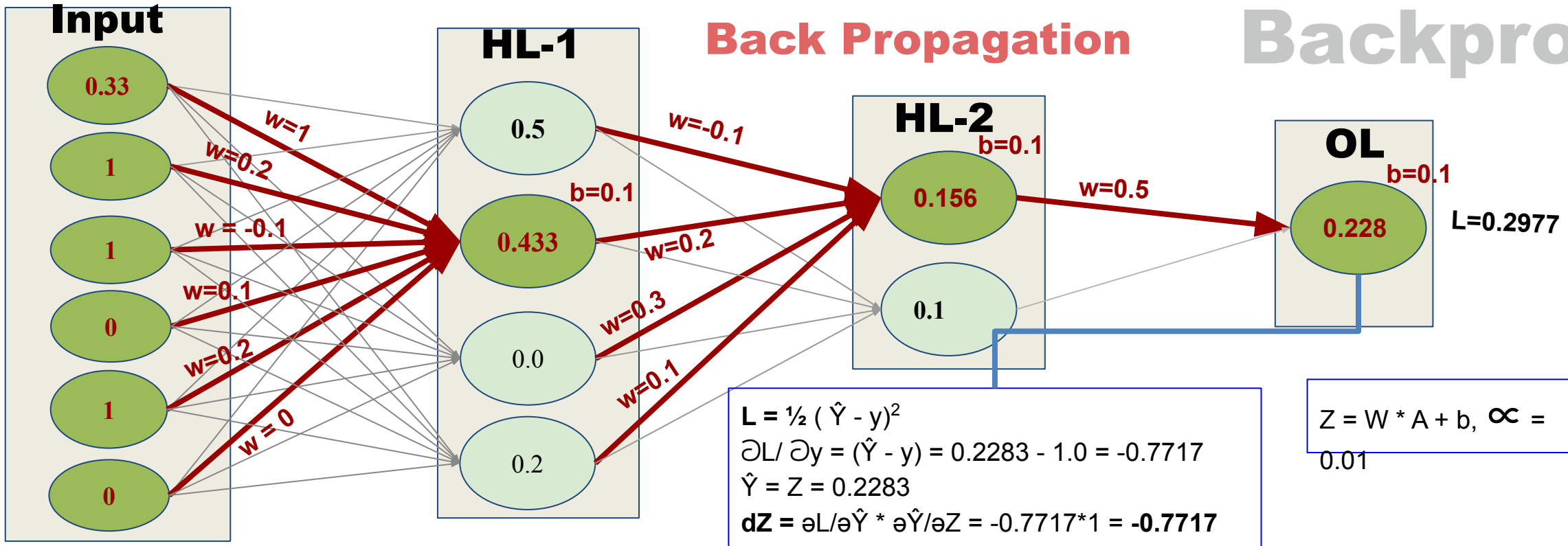
Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Back Propagation

Backprop



$L = \frac{1}{2}(\hat{Y} - y)^2$

$\partial L / \partial y = (\hat{Y} - y) = 0.2283 - 1.0 = -0.7717$

$\hat{Y} = Z = 0.2283$

dZ = $\partial L / \partial \hat{Y} * \partial \hat{Y} / \partial Z = -0.7717 * 1 =$ **-0.7717**

dW = $dZ * A = -0.7717 * 0.156 =$ **-0.1208**

db = $dZ * 1 = dZ =$ **-0.7717**

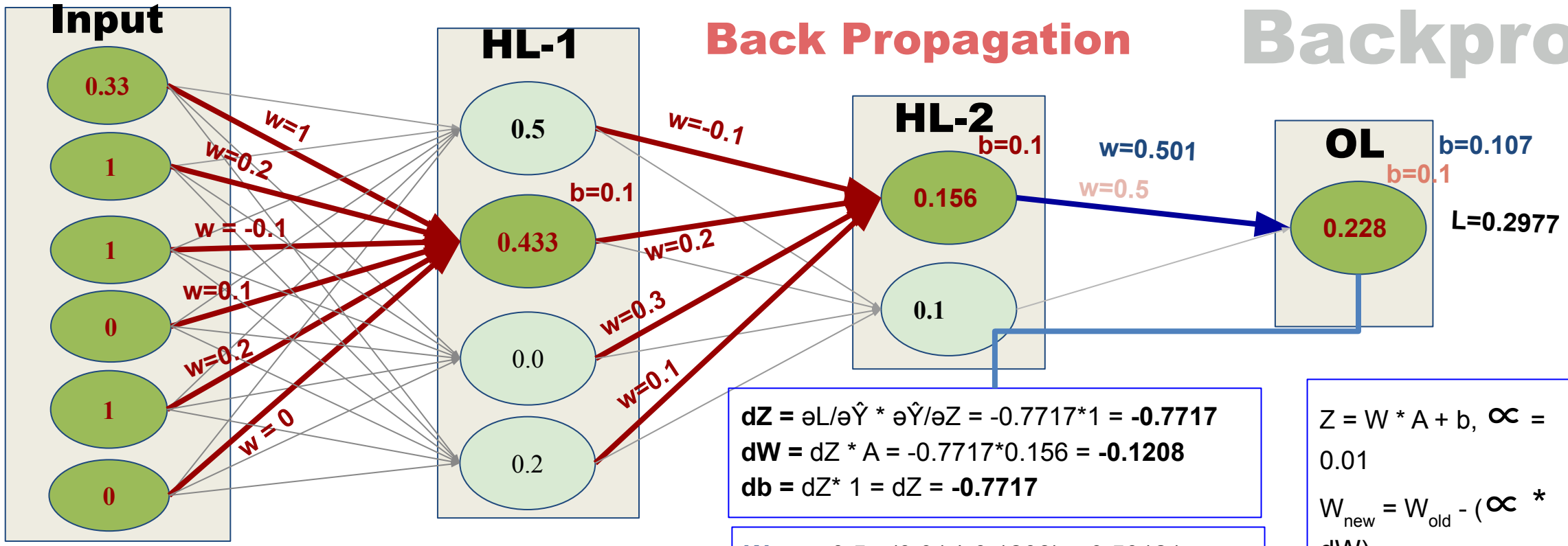
$Z = W * A + b$, $\propto$ = 0.01

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Back Propagation

Backprop

dZ = ∂L/∂Ŷ * ∂Ŷ/∂Z = -0.7717*1 = **-0.7717**
dW = dZ * A = -0.7717*0.156 = **-0.1208**
db = dZ* 1 = dZ = **-0.7717**

$Z = W * A + b$, $\propto$ = 0.01

$W_{new} = W_{old} - (\propto * dW)$

W_{new} =0.5 - (0.01 *-0.1208) = 0.50121
b_{new} = 0.1 - (0.01 * -0.7717) = 0.10771

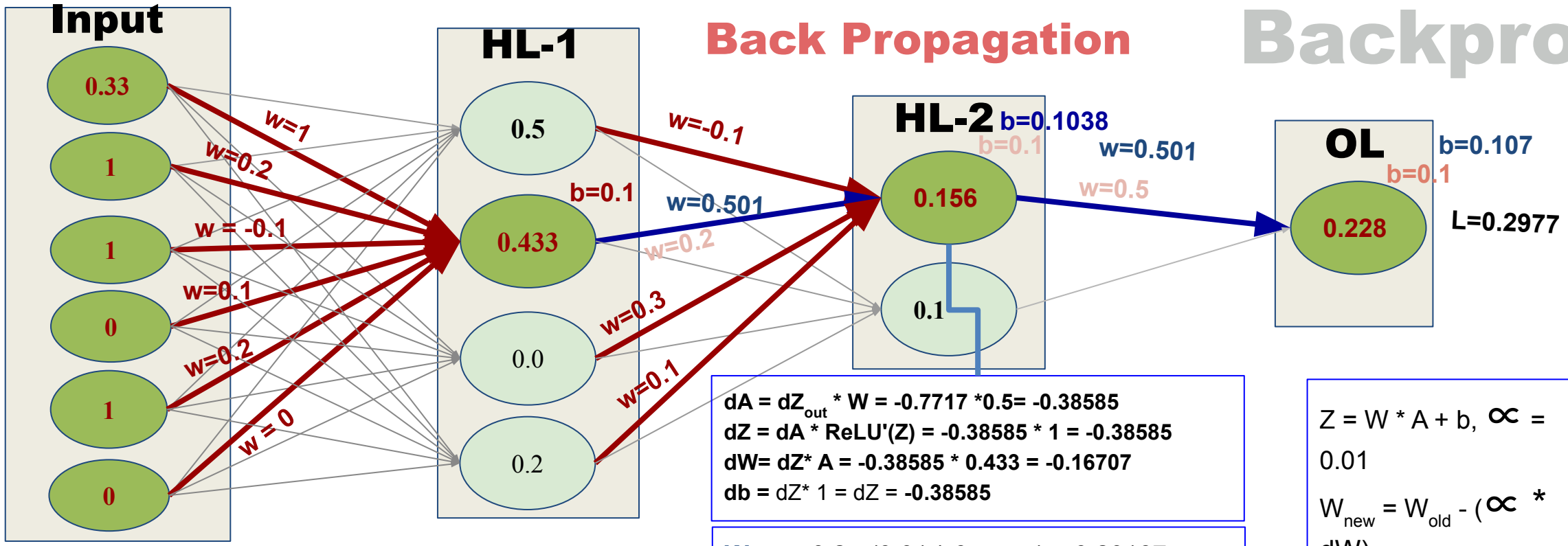
$b_{new} = b_{old} - (\propto * db)$

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Back Propagation

Backprop

$dA = dZ_{out} * W = -0.7717 * 0.5 = -0.38585$

$dZ = dA * ReLU'(Z) = -0.38585 * 1 = -0.38585$

$dW = dZ * A = -0.38585 * 0.433 = -0.16707$

$db = dZ * 1 = dZ = -0.38585$

$W_{new} = 0.2 - (0.01 * -0.16707) = 0.20167$

$b_{new} = 0.1 - (0.01 * -0.38585) = 0.10386$

$Z = W * A + b, \propto = 0.01$

$W_{new} = W_{old} - (\propto * dW)$

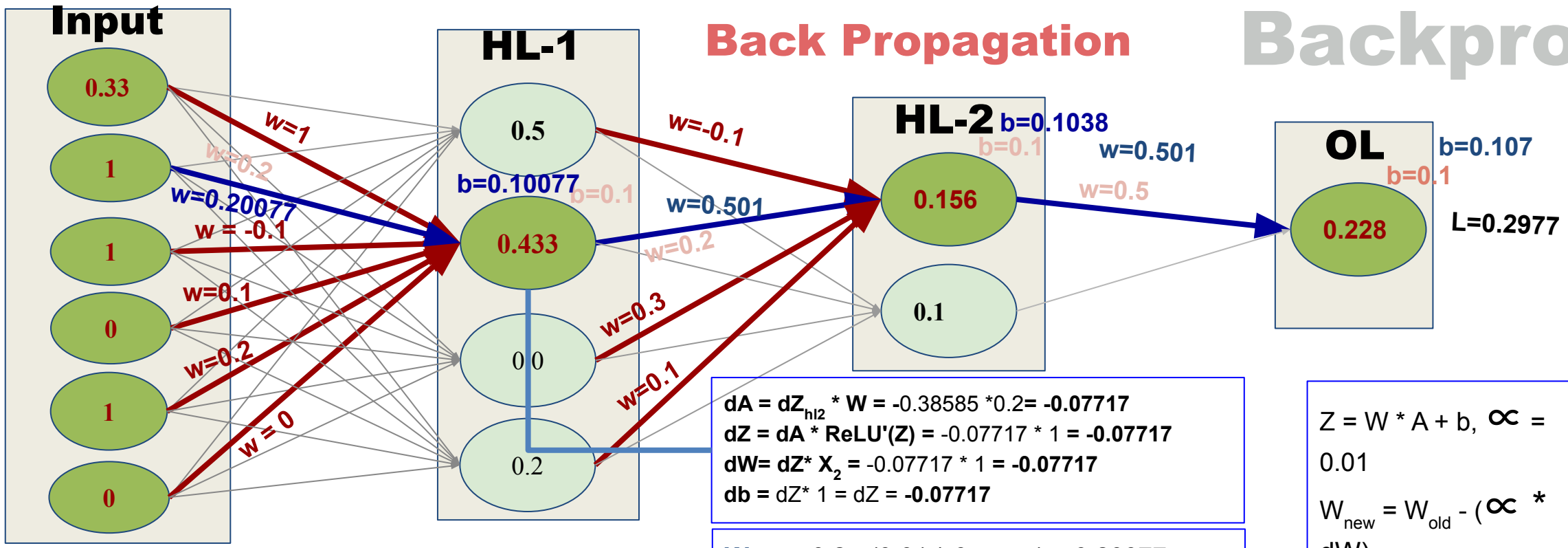
$b_{new} = b_{old} - (\propto * db)$

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Back Propagation

Backprop

$dA = dZ_{hl2} * W = -0.38585 * 0.2 = \mathbf{-0.07717}$

$dZ = dA * ReLU'(Z) = -0.07717 * 1 = \mathbf{-0.07717}$

$dW = dZ * X_2 = -0.07717 * 1 = \mathbf{-0.07717}$

$db = dZ * 1 = dZ = \mathbf{-0.07717}$

$W_{new} = 0.2 - (0.01 * -0.07717) = 0.20077$

$b_{new} = 0.1 - (0.01 * -0.07717) = 0.10077$

$Z = W * A + b, \propto = 0.01$

$W_{new} = W_{old} - (\propto * dW)$

$b_{new} = b_{old} - (\propto * db)$

Input X = [0.33, 1.0, 1.0, 0.0, 1.0, 0.0]^T

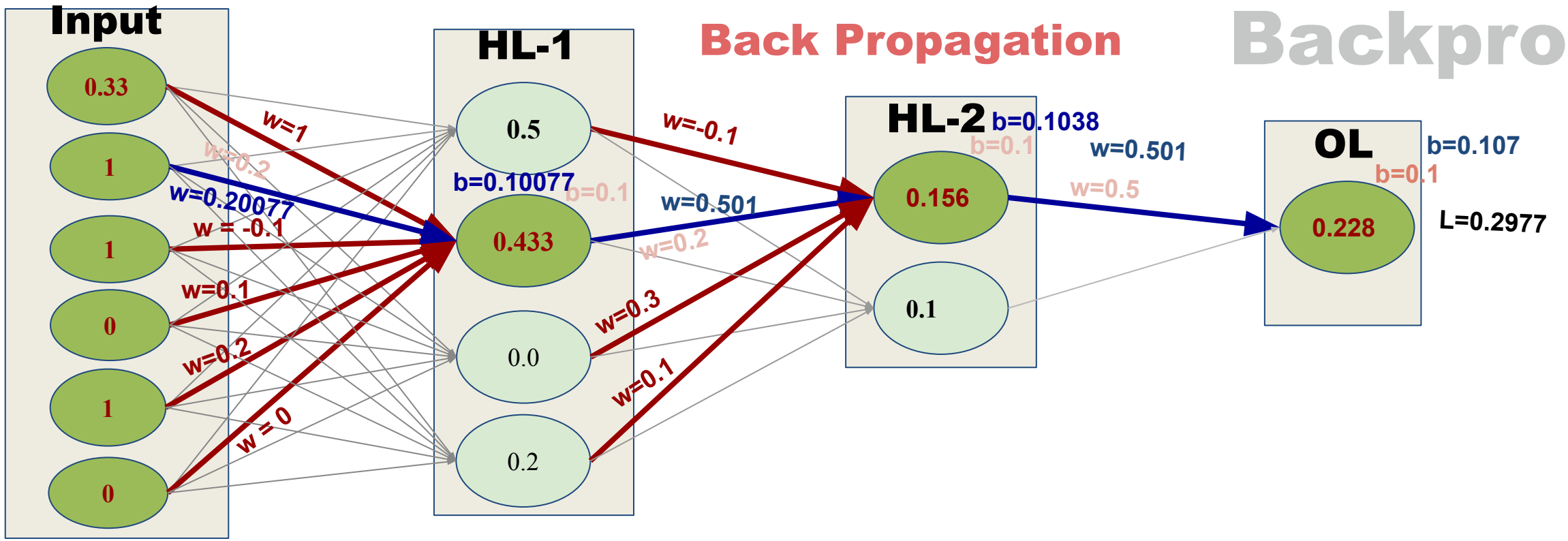
Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Back Propagation

Backprop



NN with Backprop problems

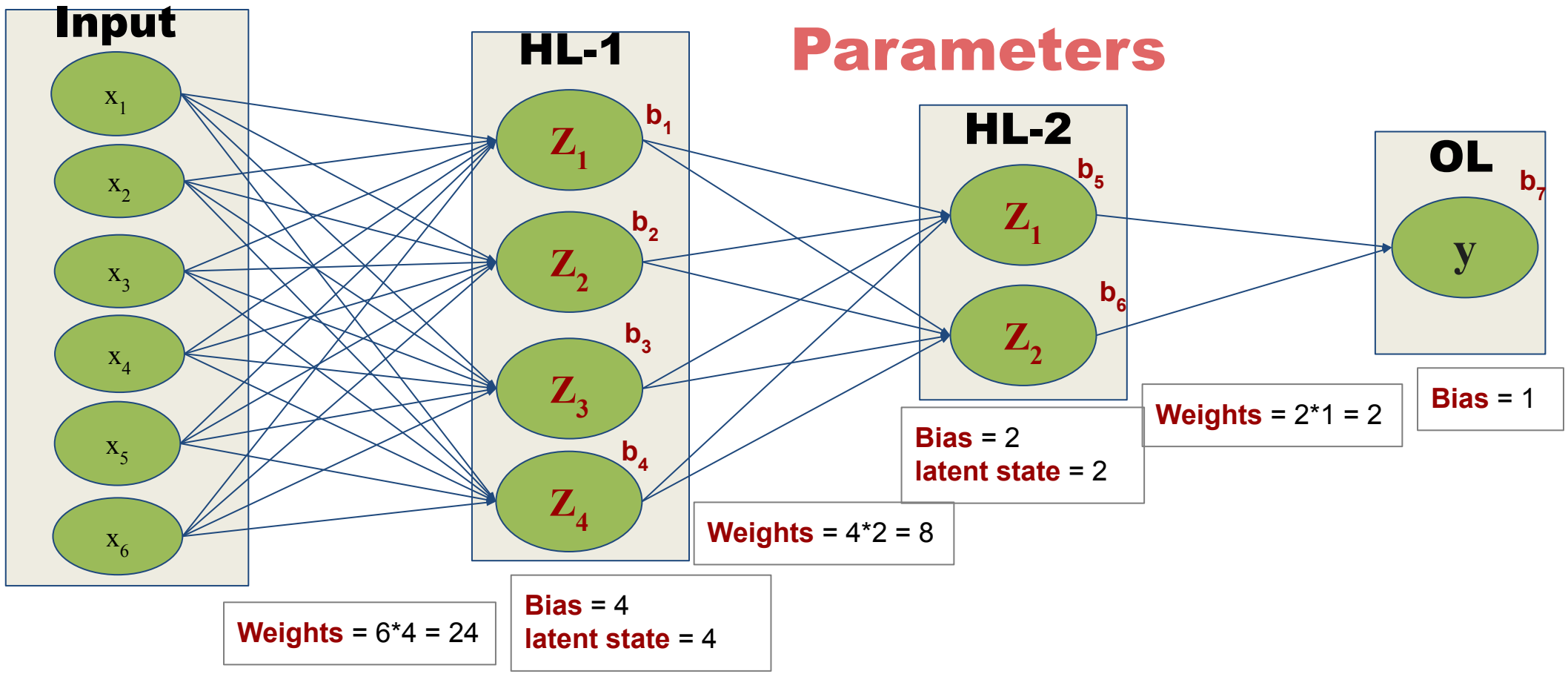
- Requires **exact weight symmetry** between forward and backward paths.
- Uses global error signals, **not local learning**.
- **Update-Locking** problem
- **Global feedback pathway problem**: a distinct error feedback pathway that starts at the output and moves backward, but this pathway does not influence neural activity — only weights.
- Is **biologically implausible** and poorly aligned with how brains learn.

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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2. Arbitrary Graph
PC Networks

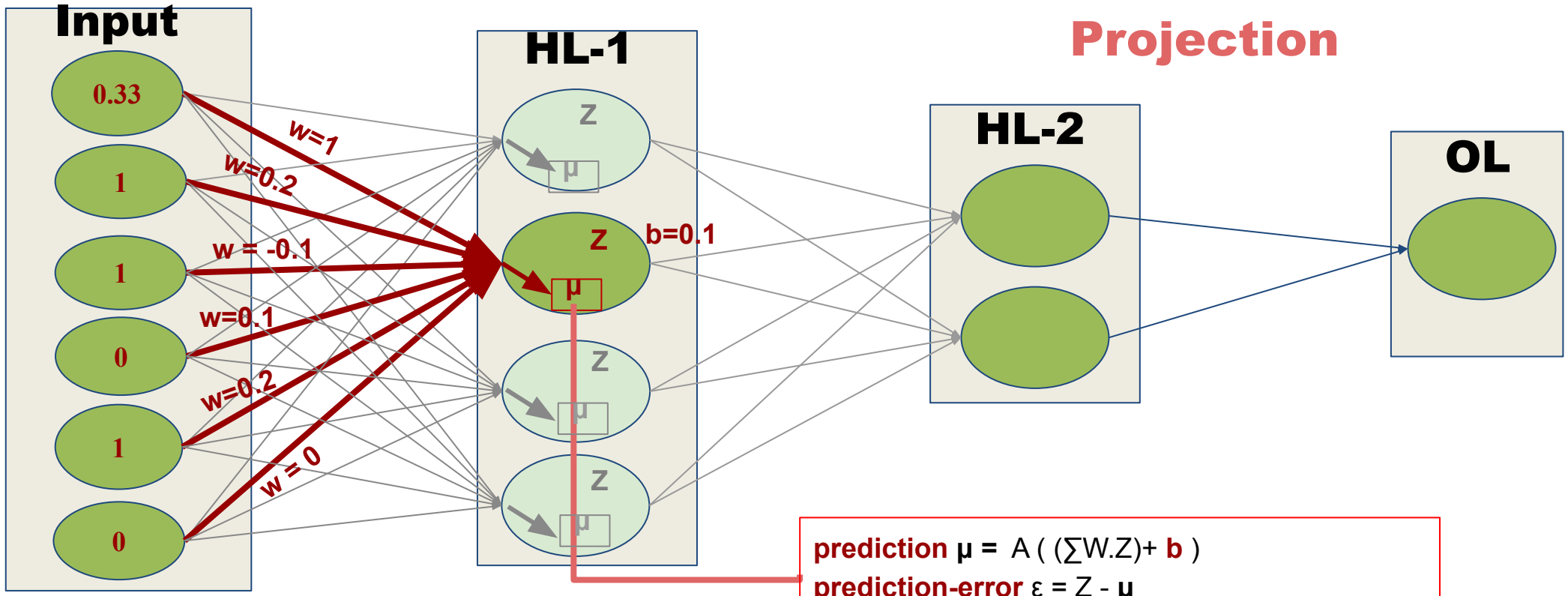
Parameters



Total Weights = Number of connections = 34
Total Bias = Number of neurons in **Hidden layers** and **output layer** = 4 + 2 + 1= 7
Total latent states = Number of neurons in **Hidden layers** = 4 + 2 = 6
Total Parameters = 34 + 7 +7 = 48

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Projection

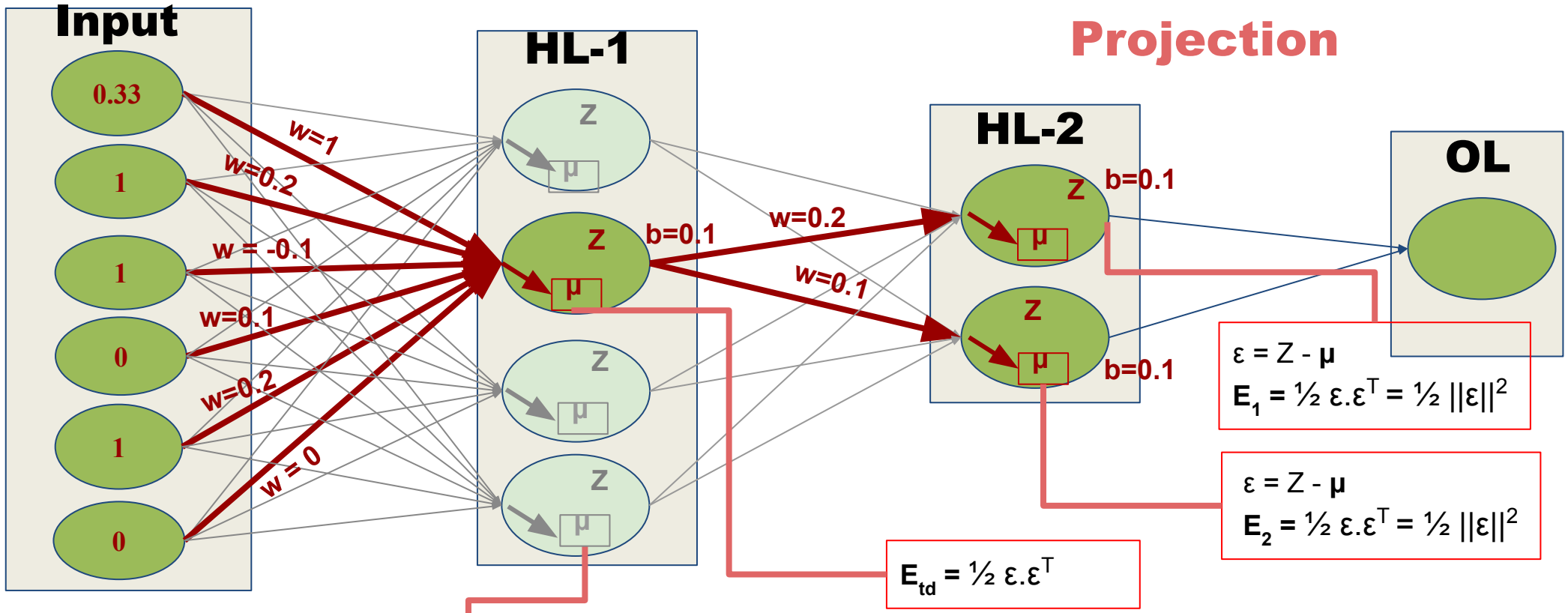
PC

prediction μ = A ((∑W.Z)+ **b**)
prediction-error ε = Z - **μ**
Energy **E** = ½ ε.ε^T

z: latent state
μ: predicted state

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Projection

PC

z: latent state
μ: predicted state
ε: prediction error
E: Energy

HL-1 Energy

$E_{pre} = E_1 + E_2$

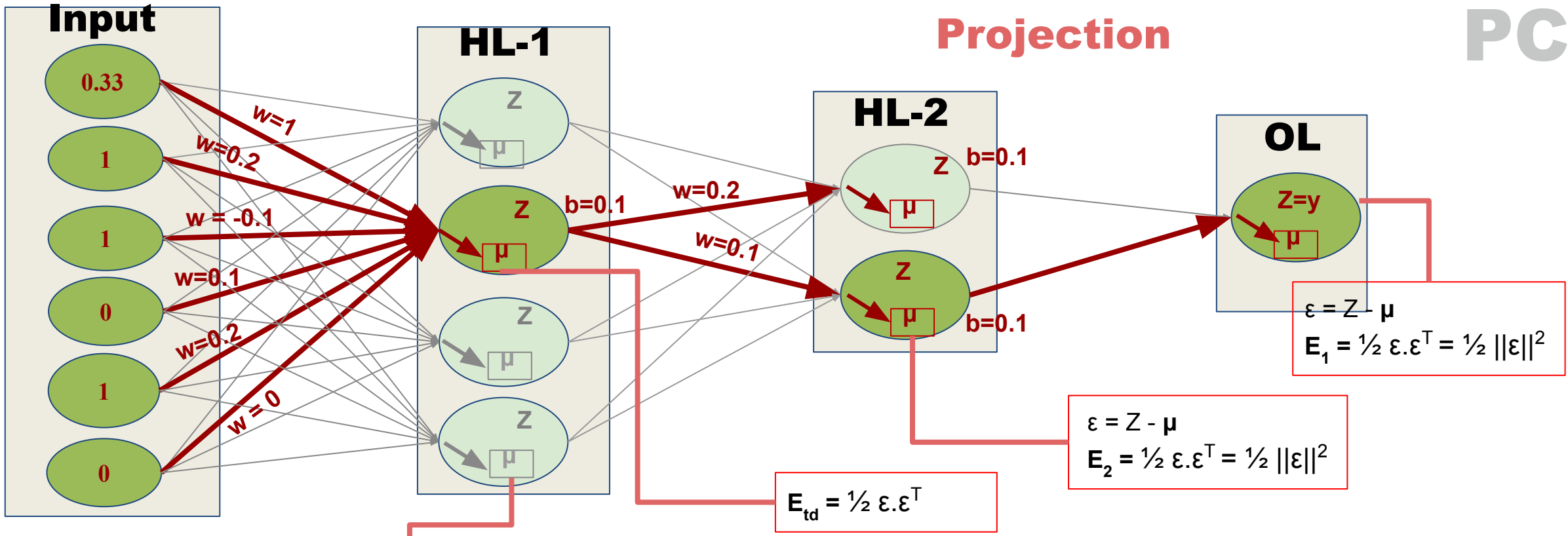
$E_{src} = E_1 + E_2 + E_3 + E_4$

$E_{total} = E_{pred} + E_{src}$

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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z: latent state
μ: predicted state
ε: prediction error
E: Energy

HL-1 Energy

$E_{pre} = E_1 + E_2$

$E_{src} = E_1 + E_2 + E_3 + E_4$

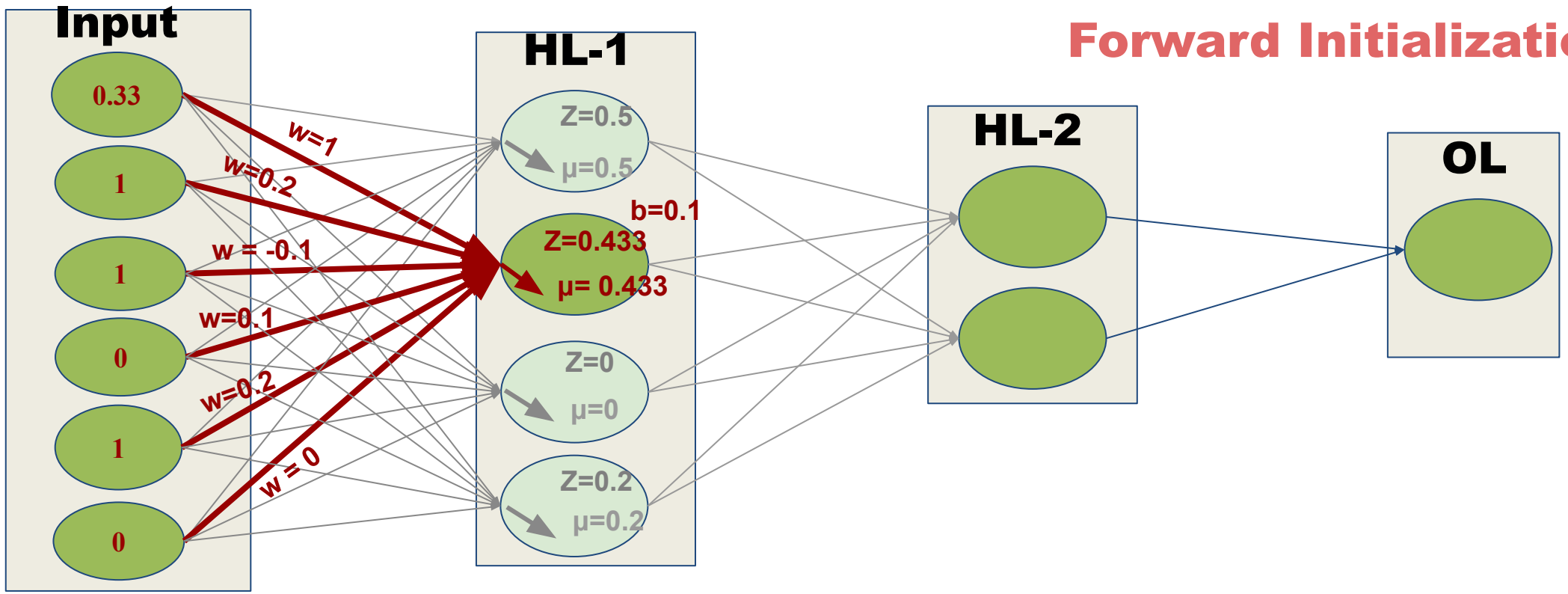
$E_{total} = E_{pred} + E_{src}$

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

Target y = 1.0

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Forward Initialization

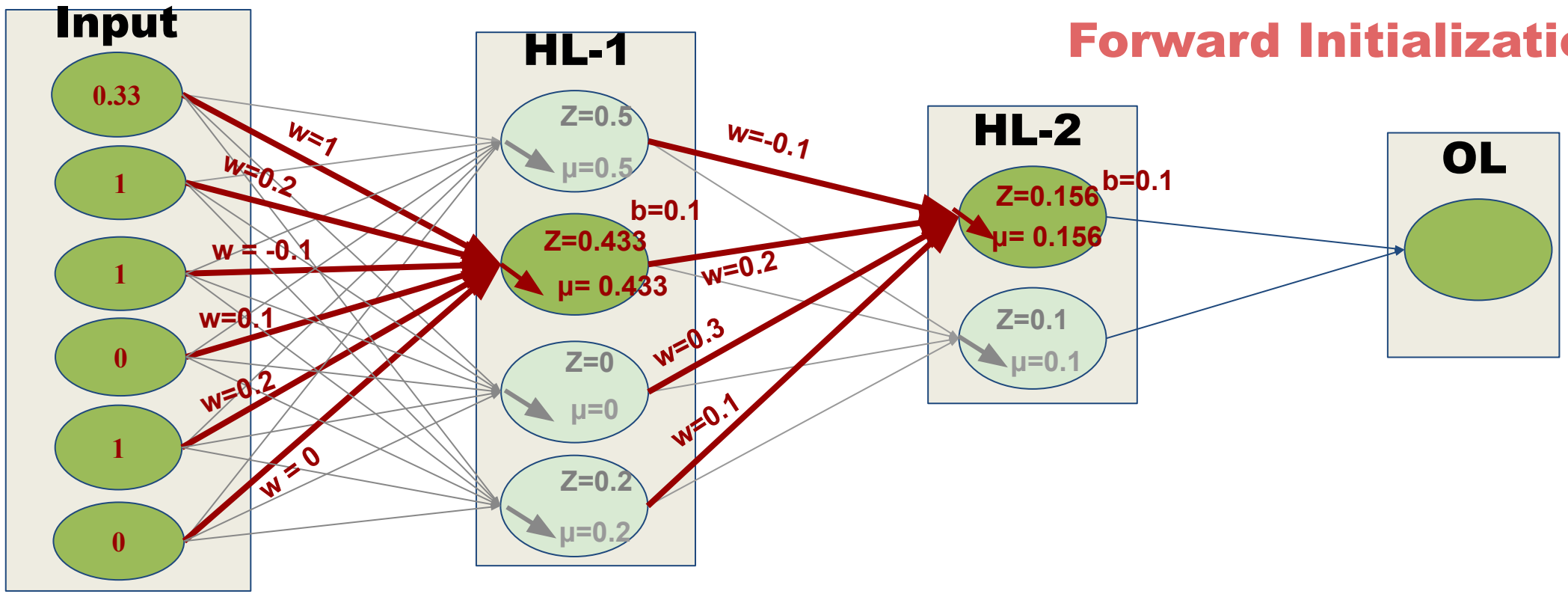


z: latent state
μ: predicted state

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Forward Initialization

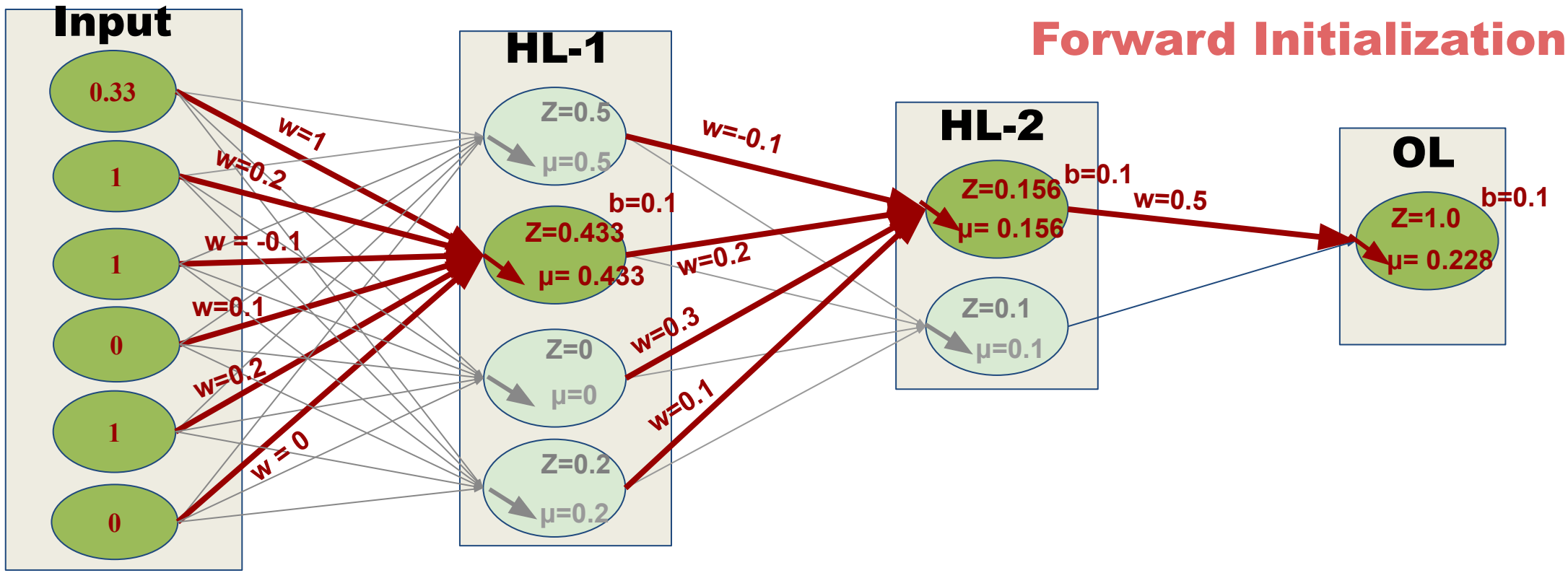


z: latent state
μ: predicted state

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Forward Initialization

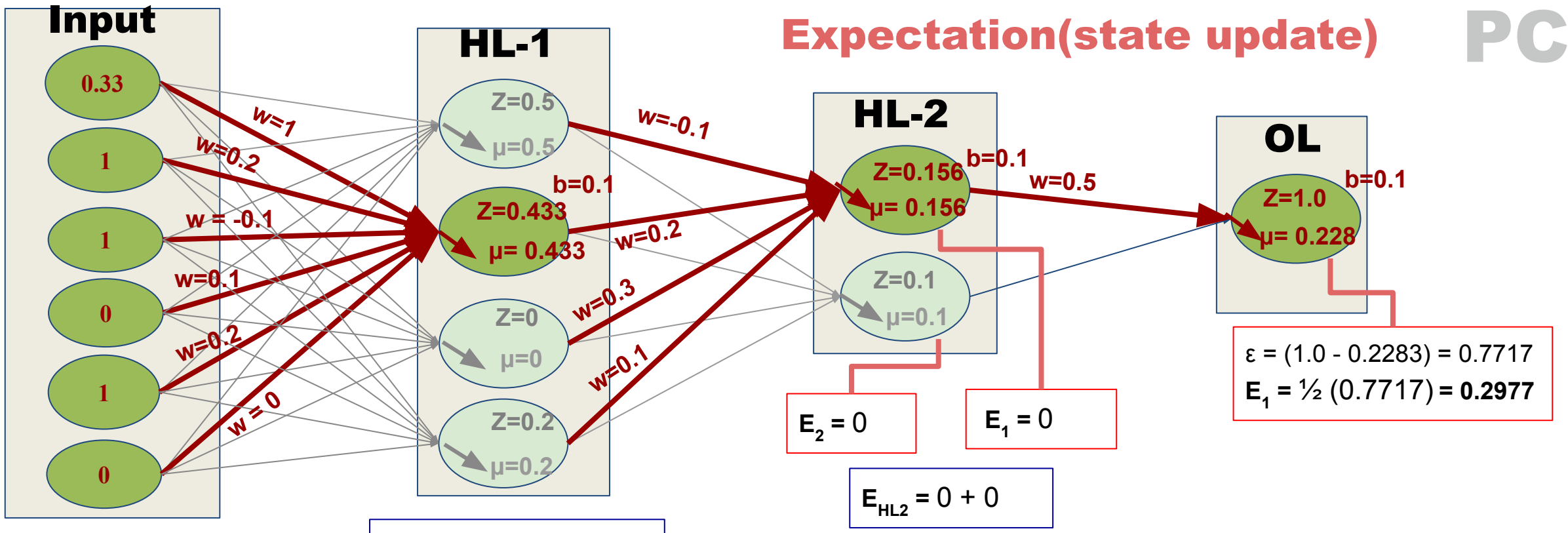


T: inference time step = **0**

step(T)	HL-1	HL-2	Output Layer
0	**z=μ**	**z=μ**	**z=μ**
1			
2			

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

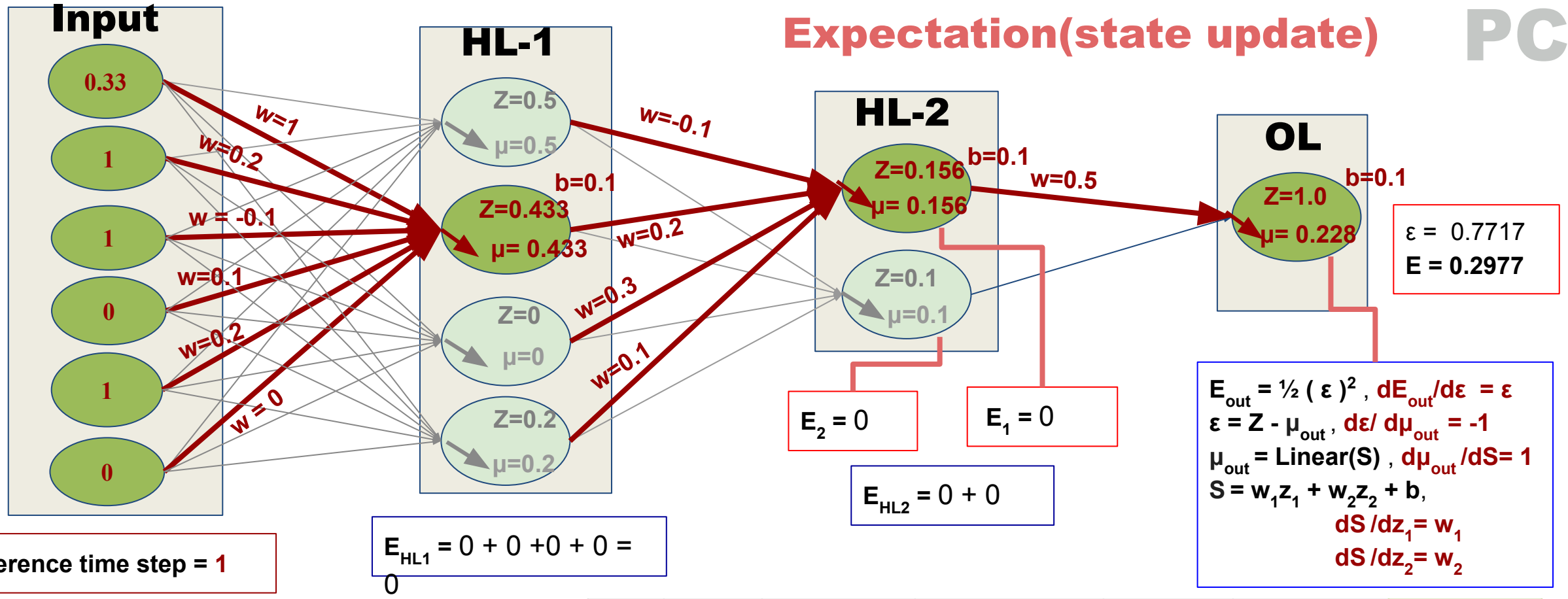
@iCogLabs



step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2			

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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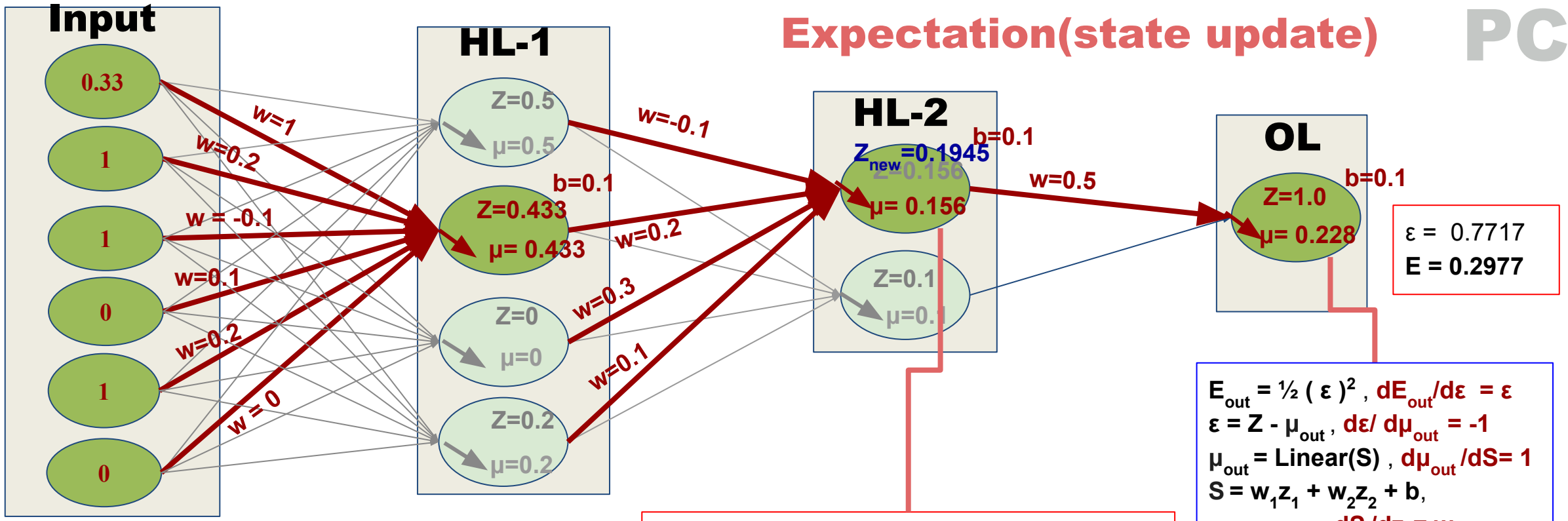
step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2			

Age	Salary	MS_Married	MS_NotMarried	Sex_Male	Sex_Female	Target Loan
0.33	1.00	1.0	0.0	1.0	0.0	1.00
0.00	0.435	0.0	1.0	0.0	1.0	0.417
1.00	0.00	1.0	0.0	0.0	1.0	0.00

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Expectation(state update)

PC



ε = 0.7717
E = 0.2977

$E_{out} = ½ (ε)^2$, $dE_{out}/dε = ε$
$ε = Z - μ_{out}$, $dε/dμ_{out} = -1$
$μ_{out}$ = Linear(S) , $dμ_{out}/dS = 1$
$S = w_1z_1 + w_2z_2 + b$,
$dS/dz_1 = w_1$
$dS/dz_2 = w_2$

T: inference time step = 1

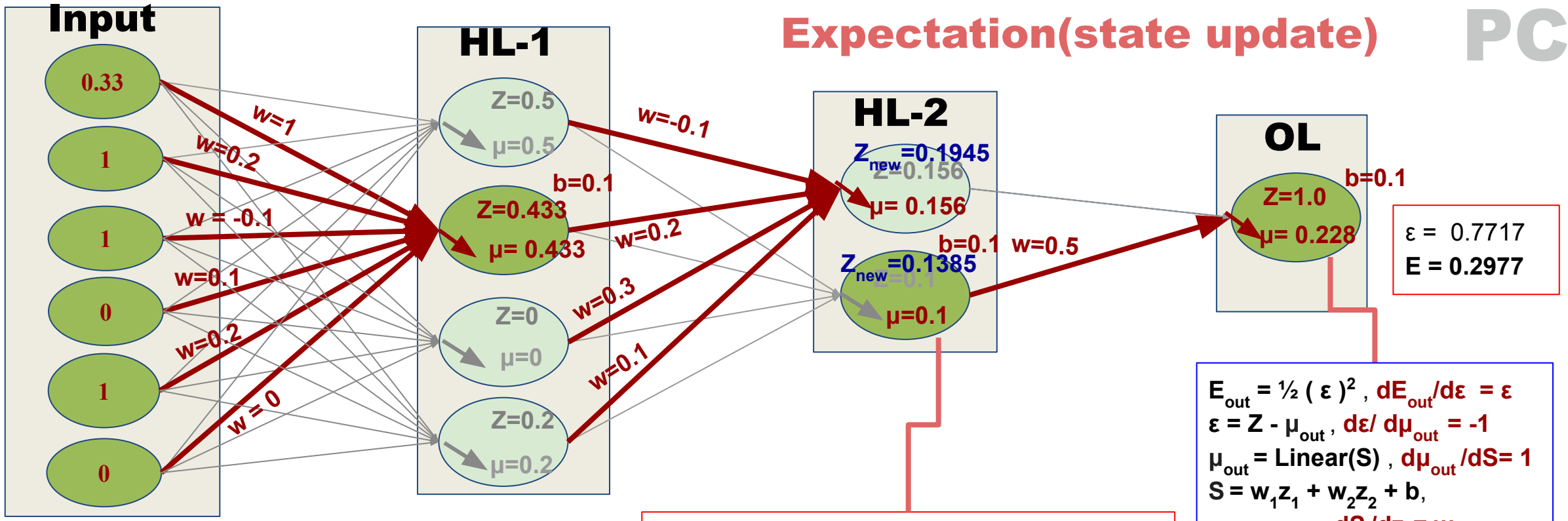
step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2			

$E_{self} = ½ (ε_1)^2 = ½ (Z_1 - μ_1)^2 = 0$
$dE_{self}/dZ_1 = (Z_1 - μ_1).(1) = ε_1 = 0$
$dE_{out}/dZ_1 = dE_{out}/dε * dε/dμ_{out} * dμ_{out}/dS * dS/dz_1$
$= ε_{out} * -1 * 1 * w_1$
$= - w_1 * ε_{out} = -0.5*0.7717 = -0.38585$
$E_{total} = E_{self} + E_{out}$
$Z_{new} = Z_{old} - η (dE_{total}/dZ_1)$
$= Z_{old} - η (dE_{self}/dZ_1 + dE_{out}/dZ_1)$
$= 0.156 - 0.1(0 - 0.38585)$
$= 0.194585$

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Expectation(state update)

PC



ε = 0.7717

E = 0.2977

T: inference time step = 1

step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2			

$E_{out} = ½ (ε)^2$, $dE_{out}/dε = ε$

$ε = Z - μ_{out}$, $dε/ dμ_{out} = -1$

$μ_{out}$ = Linear(S) , $dμ_{out}/dS = 1$

$S = w_1z_1 + w_2z_2 + b$,

$dS/dz_1 = w_1$

$dS/dz_2 = w_2$

$E_{self} = ½ (ε_2)^2 = ½ (Z_2 - μ_2)^2 = 0$

$dE_{self}/ dZ_2 = (Z_2 - μ_2).(1) = ε_2 = 0$

$dE_{out}/dZ_2 = dE_{out}/dε * dε/ dμ_{out} * dμ_{out}/dS * dS/dz_2$

$= ε_{out} * -1 * 1 * w_2$

$= - w_2 * ε_{out} = -0.5*0.7717 = -0.38585$

$E_{total} = E_{self} + E_{out}$

$Z_{new} = Z_{old} - η (dE_{total}/dZ_2)$

$= Z_{old} - η (dE_{self}/ dZ_2 + dE_{out}/dZ_2)$

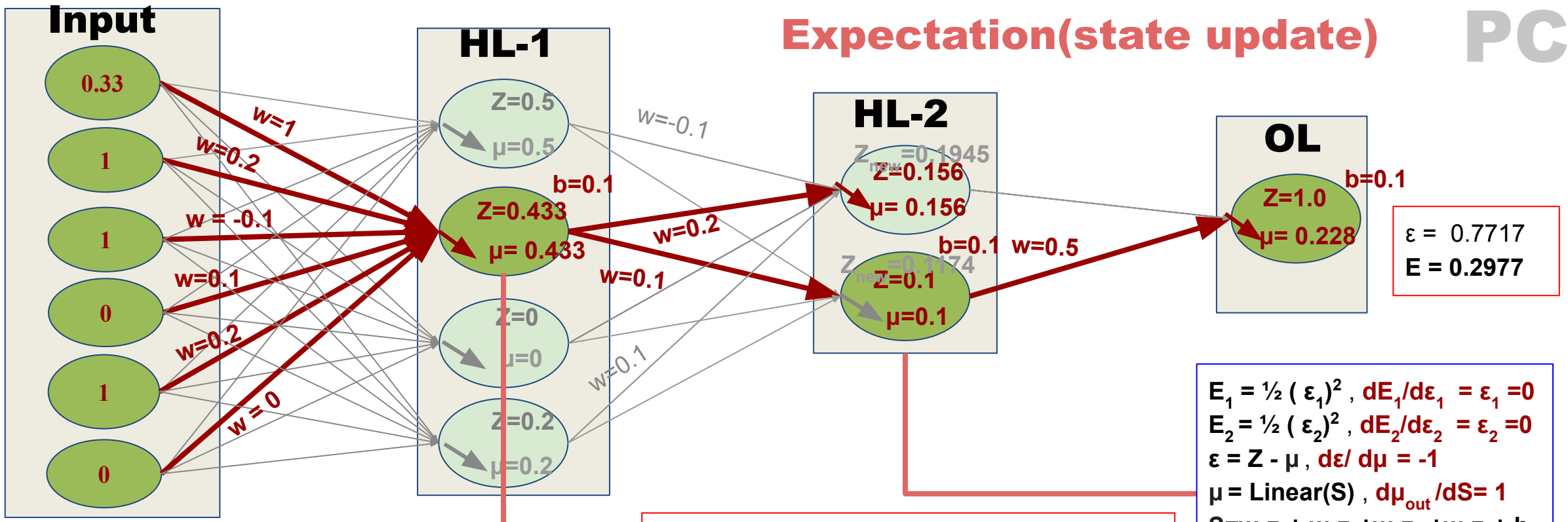
$= 0.156 - 0.1(0 - 0.38585)$

$= 0.138585$

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Expectation(state update)

PC



T: inference time step = **1**

step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2	E = 0	E = 0	E = 0

$E_1 = ½ (\varepsilon_1)^2$, $dE_1/d\varepsilon_1 = \varepsilon_1 = 0$

$E_2 = ½ (\varepsilon_2)^2$, $dE_2/d\varepsilon_2 = \varepsilon_2 = 0$

$\varepsilon = Z - \mu$, $d\varepsilon/ d\mu = -1$

μ = Linear(S) , $d\mu_{out}/dS = 1$

$S = w_1z_1 + w_2z_2 + w_3z_3 + w_4z_4 + b$,

$dS/dz_1 = w_1$

$dS/dz_2 = w_2$

$E_{self} = ½ (\varepsilon_2)^2 = ½ (Z_2 - \mu_2)^2 = 0$

$dE_{self}/ dZ_2 = (Z_2 - \mu_2).(1) = \varepsilon_2 = 0$

$dE_{HL2}/dZ_2 = dE_{HL2}/d\varepsilon . d\varepsilon/ d\mu_{HL2} * d\mu_{HL2}/dS * dS/dz_2$

$= \varepsilon_{HL2} * -1 * 1 * w_2$

$= - w_2 * \varepsilon_{HL2} = (-0.2*0) + (- 0.1*0) = 0$

$E_{total} = E_{self} + E_{out}$

$Z_{new} = Z_{old} - \eta (dE_{total}/dZ_2)$

$= Z_{old} - \eta (dE_{self}/ dZ_2 + dE_{HL2}/dZ_2)$

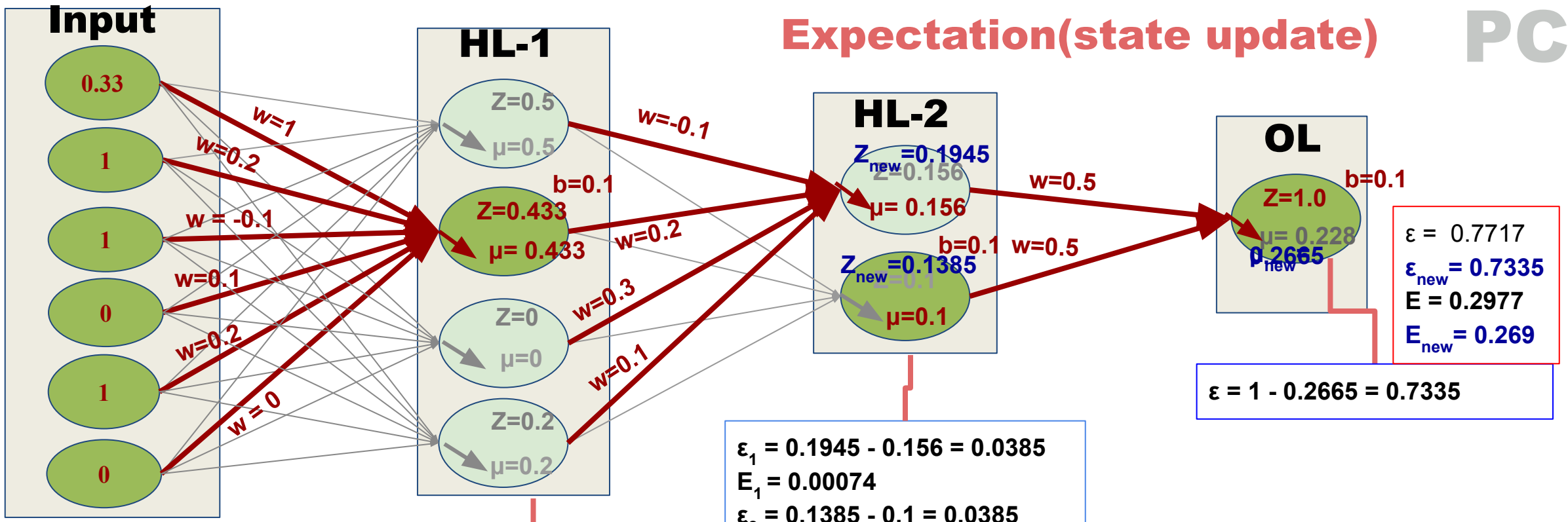
$= 0.433 - 0.1(0 + 0)$

$= Z_{old}$

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Expectation(state update)

PC



T: inference time step = **2**

step(T)	HL-1	HL-2	Output Layer
0	z=μ	z=μ	z=μ
1	E = 0	E = 0	E= 0.2977
2	E = 0	E = 0.0015	E= 0.269

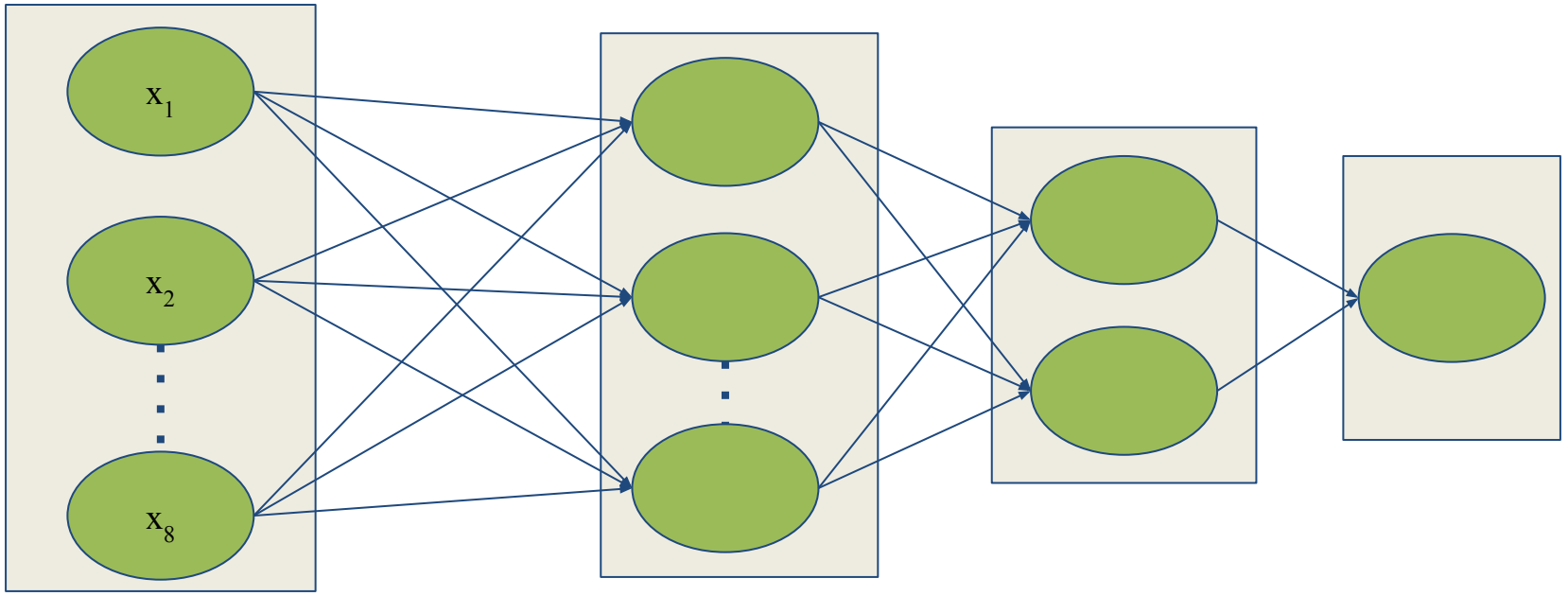
at each inference time step (**T**)

- predict neuron activity (**μ**)
- calculate prediction error(**ε**)
- update state (**Z**)

$$Z_{new} = Z_{old} - \eta\,(dE_{total}/dZ)$$
$$= Z_{old} - \eta\,(dE_{self}/dZ + dE_{next_layer}/dZ)$$
$$W_{new} = W_{old} - \propto (dE/dW)$$

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Backprop

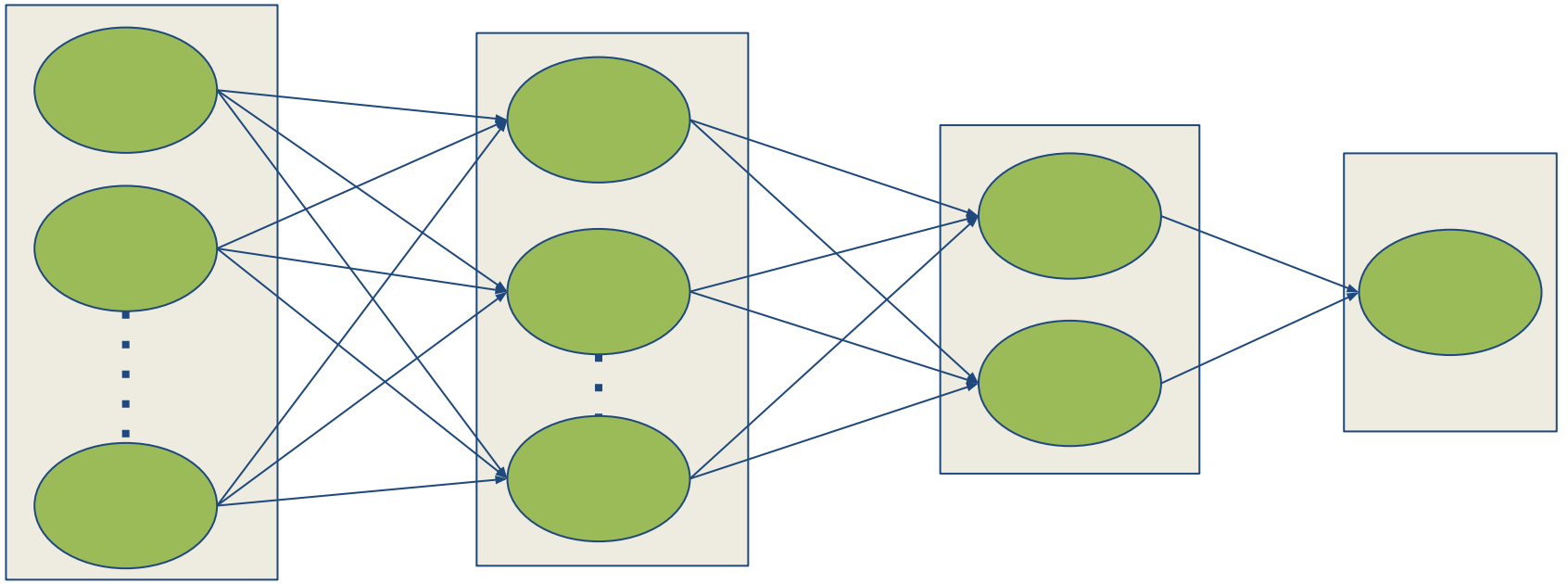


- Strict feedforward pass for activations
- **Information Flow:** Sequential (Top-down predictions, bottom-up errors)
- **Error Calculation:** Global (Computed only at the output layer)
- **Learning Mechanism:** Chain rule of calculus distributing global error
- **Biological Plausibility:** Low (The brain doesn't pause to run a backward pass)

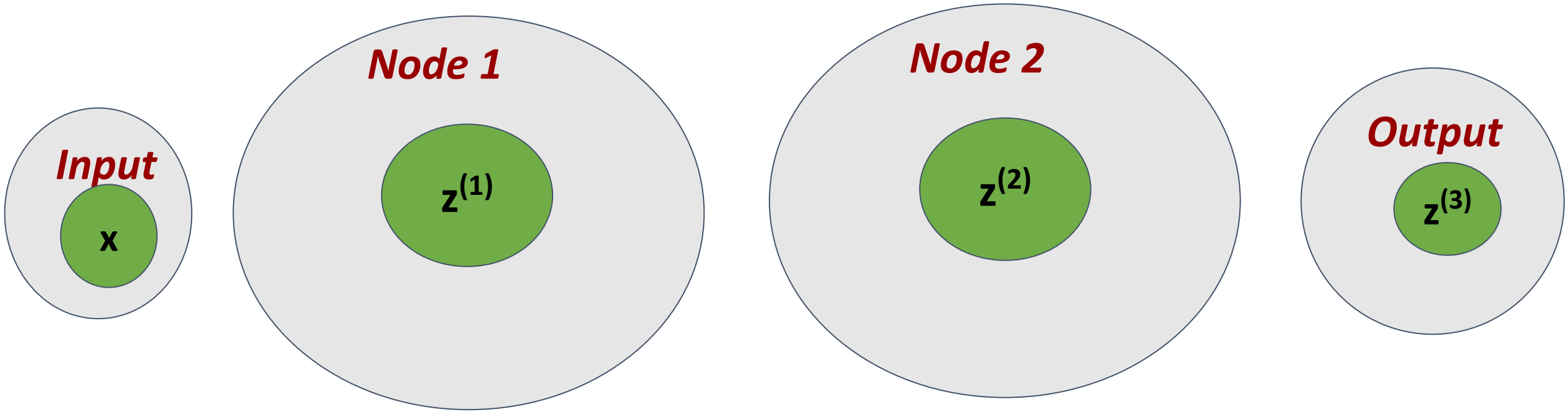
Vs

PC

- Hidden layer neurons have states
- **Information Flow:** Bidirectional (Top-down predictions, bottom-up errors)
- **Error Calculation:** Local (Computed at every layer simultaneously)
- **Learning Mechanism:** Minimizing local prediction errors in real-time
- **Biological Plausibility:** High (Mimics human cortical architecture)



Predictive Coding Architecture



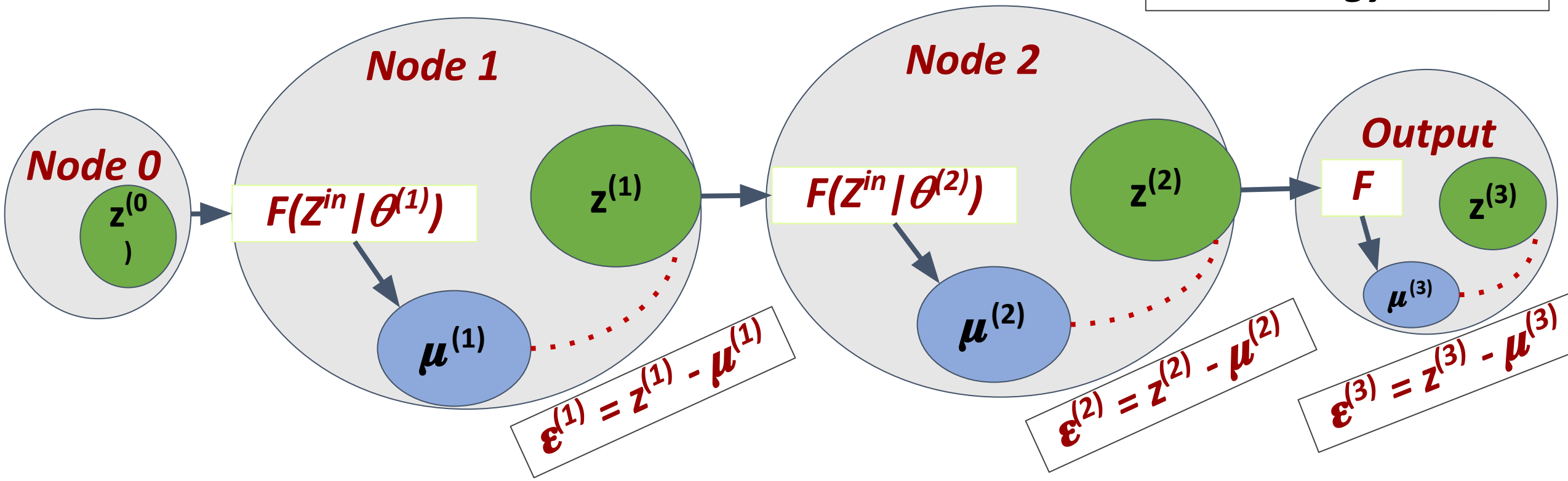
x = input data
$z^{(1)}$, $z^{(2)}$ = latent / hidden states
z(4) = output state

input clue → network searches memory → returns closest remembered pattern

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Predictive Coding Architecture

Prediction Error:ε
Energy: E



$$E = (\varepsilon^{(1)})^T . \varepsilon^{(2)}$$

$$z^{(1)} = z^{(1)} - \eta(\nabla_z E^{(1)} + \nabla_z E^{(2)})$$

$$W^{(1)} = W^{(1)} - \propto(\nabla_w E^{(2)})$$

Each node has

- z = actual state / latent state
- μ = predicted state / projection
- ε = prediction error
- θ = parameters;W, b = weights, biases,
- F = transfer / activation function
- E = energy functional

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Training Task

Task Submission Requirements:

1. **Standard Neural Network (Backpropagation**
   - Implement a neural network architecture trained using standard backpropagation.
2. **Predictive Coding (PC) Architecture**
   - Implement a Predictive Coding architecture, ensuring you clearly demonstrate:
     - Layer-wise energy.
     - A solid understanding and implementation of:
       - Layer-wise prediction error calculations.
       - Latent state updates.
       - Total system energy.
3. **Comparative Analysis**
   - Compare both architectures (Backpropagation vs. Predictive Coding) and their results.
4. **Discussion**
   - Explain what limitations of standard backpropagation are addressed by Predictive Coding.